

Computational Modeling for Psychology Students

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1 Identifying Details

PSYCH420 Section 001

Meets: Friday 8:30-11:20

Room: EV3

2 What will we study and how will we study it?

1. What is the difference between a mind and a brain?

2. What is the difference between computational psychology and computational neuroscience?
3. Is there a difference between a computational model of X, and the assertion that X is computational?
4. What is computation?
5. What is cognition?
6. What does it mean to assert that cognition is computation?

3 A word about the use of LLMs

3.1 What are Marr's three levels and how do they relate to the question of whether cognition is computational?

David Marr proposed three levels of analysis for understanding cognitive systems:

1. Computational Level: Defines the goal or problem the cognitive system is solving and the logic/strategy for solving it. For cognition, this means specifying what information processing task is being accomplished.
2. Algorithmic Level: Describes the specific representation and algorithm used to implement the computational goal. Here, one details the precise steps and data structures used to achieve the computational objective.
3. Implementation Level: Examines the physical mechanisms that actually realize the algorithm, such as neural structures in the brain.

Regarding computational cognition, Marr's levels suggest cognition can be understood computationally: If cognitive processes can be systematically described at the computational and algorithmic levels, with clear input-output mappings and step-by-step information processing, this supports viewing cognition as fundamentally computational. The implementation level reveals how these computational processes are physically instantiated.

By providing a structured framework for analyzing cognitive systems, Marr's levels offer a strong argument for computational

approaches to understanding mental phenomena, while also acknowledging the importance of examining actual biological mechanisms.

Claude_{AI} <2024-11-25 Mon>

4 For next week

4.1 Reading

https://pmc.ncbi.nlm.nih.gov/articles/PMC3006465/pdf/10867_2010_Article_9195.pdf

4.2 Expectations

1. You will make some effort to look-up things you do not understand.
2. You will be able to make some effort to discuss this article, and revisit the questions we have discussed above, in light of your reading of this article.
3. Quizzes are an imprecise way of assessing your learning, but in the absence of discussion I have no way of gauging what you know, and so I can make five question quizzes on the readings if necessary. Please help me not to have to do this.

4.3 You will turn in

1. a less than one page paper in which you indicate the poster/talk you have selected from here and why you chose this is an example of a computational approach to a psychological question, and why it resonated with you.
2. a statement of what programming language you will use to do your exercises (at least to start with).
 - (a) If you are already a pretty accomplished programmer pick something new to challenge yourself with and learn a bit more: common lisp; scheme; racket; haskell; ocaml. Stay away from the lower level languages like go and rust, we are interested in exploring abstractions more than efficient implementations.

5 Technical Expectations

1. You need a computer. Phones and tablets will not work.
2. You need to know how to use your computer:
 - can you find a particular file or directory?
 - do you know how to download a program or data and find it on your computer?
 - do you know how to use "git" or any version control software?
 - do you know what an IDE is and can you use it for the basics? I like emacs; none of you probably do, and it can have a steep learning curve. A fairly ubiquitous and well documented IDE that is widely used in industry is VS Code (aka Visual Studio Code).
 - Can you program at a rudimentary language in any programming language. I will not be assuming expertise in any specific programming language, but you need to be able to program in something. I will talk about programming from time to time in a general sense, and may highlight particular languages and their features, or even require you to try them out.
3. You need to bring your computer to class. There will be many occasions on which class time will be used for you to work on a computing project, and then we will discuss it in class or you will submit a more extensive exploration as a homework assignment.

6 And if we have some time left

```
def xcubed (x) : return (x * x * x)
def diff_x_cubed (x) : return (3 * x * x)
def get_step (guess, goal) :
    return ((goal - (xcubed (guess))) / (diff_x_cubed (guess)))
def get_cube_root (goal, initial_guess, tolerance) :
    cur_guess = initial_guess + get_step(initial_guess, goal)
    error = abs(xcubed(cur_guess) - goal)
    while (error > tolerance):
        cur_guess = cur_guess + get_step(cur_guess, goal)
        error = abs(xcubed(cur_guess) - goal)
    return(cur_guess)
```